KING SAUD UNIVERSITY
FACULTY OF APPLIED COMPUTER SCIENCES
COURSE: **cos370/372**

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Network Switch Automation Using GNS3 and Python

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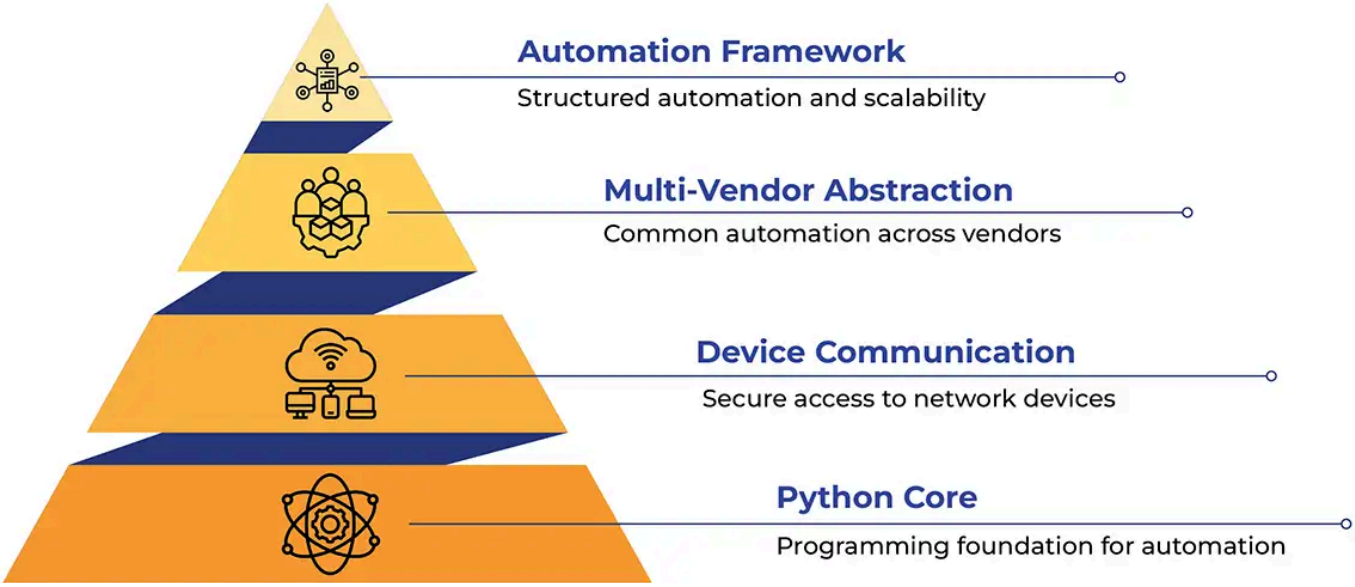
1.1 Introduction

With the rapid growth of modern networks, manual configuration of devices such as switches and routers has become inefficient and prone to errors. Network automation addresses this issue by using programming tools like Python and libraries such as Netmiko to remotely configure devices over secure connections (SSH over TCP port 22).

In this project, an automated network is designed and implemented using GNS3, where VLANs are configured at the Data Link layer to segment the network logically. The automation scripts simplify repetitive tasks, improve accuracy, and demonstrate how networking concepts can be integrated with software to achieve efficient and scalable network management.



1.2 Objectives

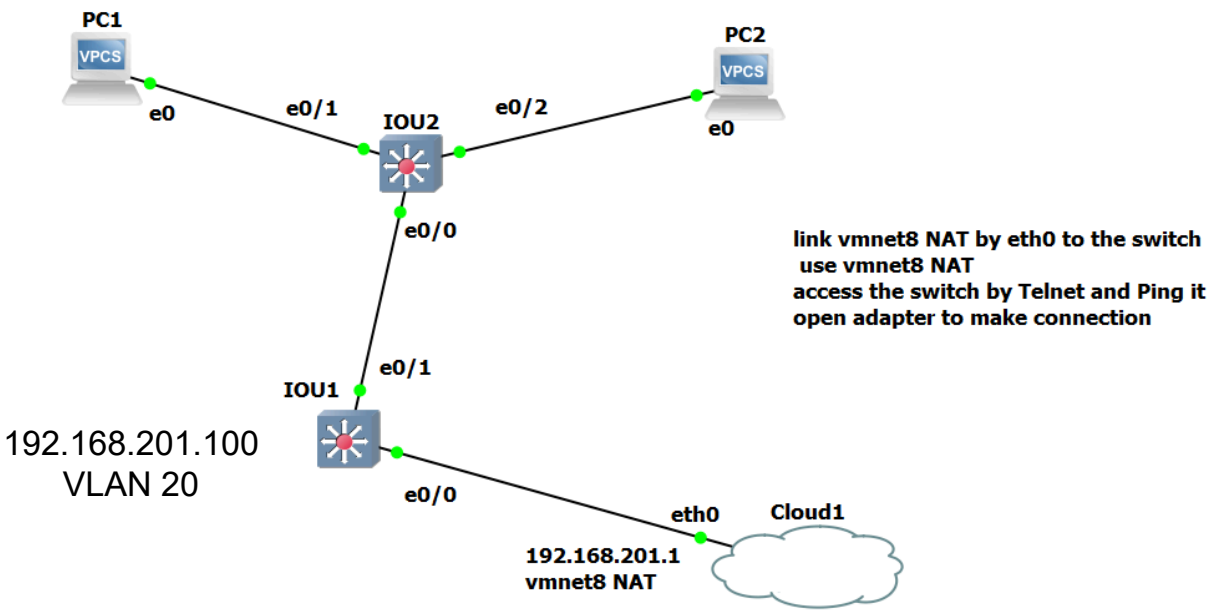
-Automate switch confiuration

-Implement VLAN Segmentation

-Reduce Manual Errors

-Improve Scalability

1.3 Network Topology



1.4 VLAN Trunking Frame

Md | Ms | type/length | TAG

TAG

```
0000  00 0c 29 4d b2 8b 00 0c 29 73 73 db 81 00 81 00
0010  00 28 45 00 00 3c 00 b7 00 00 80 01 c1 5d b2 0f
0020  0a 45 b2 0f 0a 49 08 00 4d 56 00 01 00 05 61 62
0030  63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72
0040  73 74 75 76 77 61 62 63 64 65 66 67 68 69
```

Md	Ms	type/length		
00 0c 29 4d b2 8b	00 0c 29 73 73 db	81 00	DATA	FCS

Md	Ms	type/length			
00 0c 29 4d b2 8b	00 0c 29 73 73 db	TAG	81 00	DATA	FCS

type	Pri	CFI	VID=40
0x8100	000	0	0000 0010 1000
2B	3b	1b	12b

1.5 Testing And Verification

connectivity from Host to Switch VLAN 20

```
C:\Users\hussi>ping 192.168.201.100

Pinging 192.168.201.100 with 32 bytes of data:
Reply from 192.168.201.100: bytes=32 time=1ms TTL=255
Reply from 192.168.201.100: bytes=32 time=1ms TTL=255
Reply from 192.168.201.100: bytes=32 time=1ms TTL=255
Reply from 192.168.201.100: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.201.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

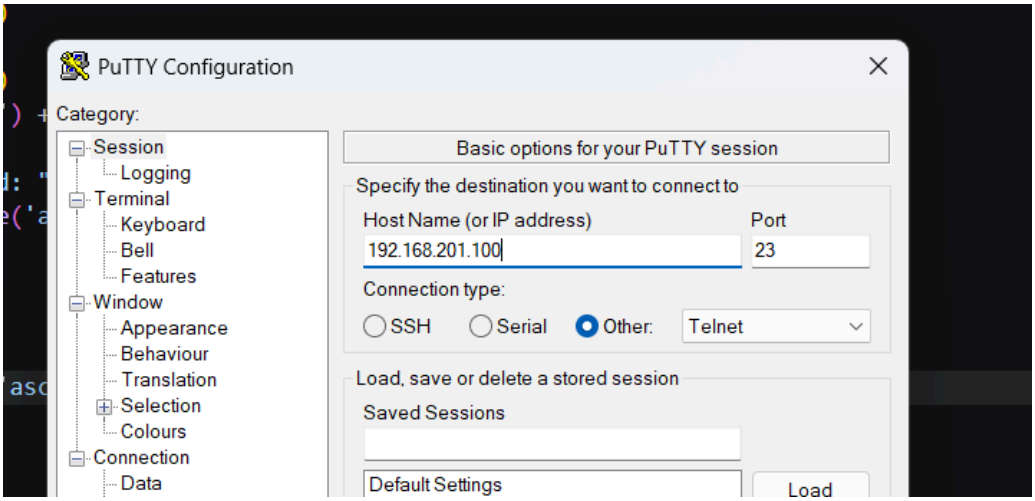
VLAN 20 and 40 (created by Python script)

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------
1    default                          active    Et0/1,
                                                Et1/1,
                                                Et2/1,
                                                Et3/1,
20   VLAN0020                         active    Et0/0
40   Management                       active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```

```
!
interface Vlan20
 ip address 192.168.201.100 255.255.255.0
!
```

1.5 Testing And Verification

Access switch through Telnet Protocol



Testing script and output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\hussi\Documents\Projects\Network Automation> &
Network Automation/Switch.py"
Username: admin
Password:
Enable Password:
New Hostname: SW1
MOTD Message: access is allowed
VLAN ID: 40
VLAN Name: Management
Trunk Port: e0/1
Access Port: e0/1

==================================================
Connecting to 192.168.201.100...
==================================================
```

Result

```
Compressed configuration from 1534 bytes to 933 bytes[OK]
IOU1(config)#
IOU1(config)#
*Apr 15 14:28:16.728: %SYS-5-CONFIG_I: Configured from console by admin on vty1 (192.168.201.1)
IOU1(config)#
S1(config)#
S1(config)#
```

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